



1 Hydrogen As A Fuel For Next-Gen Mobility

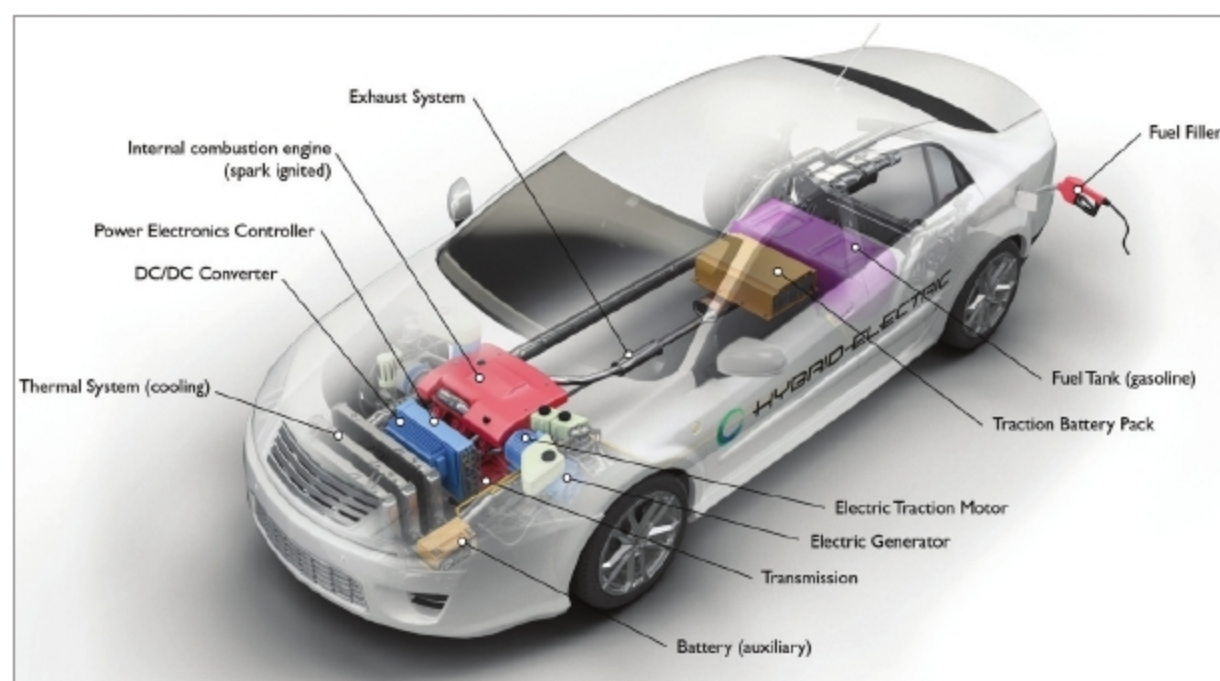
2 Wind-Solar Hybrid Systems To Improve Grid Stability

3 Innovations Thriving In The Battle Against Coronavirus

4 Automated Food Cooking

1 HYDROGEN AS A FUEL FOR NEXT-GEN MOBILITY

Hydrogen-powered vehicles are more efficient than a combustion engine as no greenhouse gases are emitted. However, the process of generating hydrogen should also be green. Commonly used processes for separating hydrogen from other substances include steam reforming of methane, gasification, and electrolysis



(Credit: <https://afdc.energy.gov>)

Hydrogen has been used as a source of clean energy throughout history, from spacecraft and submarines to refineries and forklifts. When vehicles are powered by hydrogen fuel cells, the only byproduct is water, which is emitted in the form of pure water vapour, and hence no greenhouse gases are emitted. In addition to hydrogen's abundance in nature (as part of water and atmospheric gas), hydrogen-powered vehicles are more efficient than a combustion engine since there are no moving parts.

For zero net emissions, the process of generating hydrogen should also be green. Commonly used processes for separating hydrogen from other substances include steam reforming

of methane, gasification, and electrolysis. Hydrogen is also a byproduct of many industrial processes where it is often treated as waste.

In the most popular process known as electrolysis, an electric current is passed through water to separate its primary constituents—hydrogen and oxygen. Electricity is not generated from fossil fuels but produced from sources like wind, solar, and hydro energies to mitigate carbon footprints.

Hydrogen binds to most other substances and is a flammable gas. So, it is stored in liquid form in a fuel tank for ease of transportation until it is needed by the fuel cell stack (collection of electrodes). The amount of energy stored is decided by the size of the hydrogen fuel tank. Besides,

in gaseous form, costs for compression, cooling, transport, and storage of hydrogen differ. Other key components of a hydrogen vehicle include electric traction motor(s), thermal system, auxiliary battery, traction battery pack, among others.

Manufacturers define the power of the vehicle by the size of the electric motor. Since the hydrogen engines are powered by an electric motor, these vehicles are also known as fuel cell electric vehicles (FCEVs). Power is generated when hydrogen inside fuel cells in vehicles undergoes reverse electrolysis. In this process, hydrogen present in tanks reacts with oxygen in the air to produce electrical energy, water, and heat. The electricity thus produced can either directly flow to the electric motor, which in turn is converted to mechanical energy to drive the vehicle's wheels, or the energy can be stored in the battery pack, depending on the requirement. The auxiliary battery is mainly present to start the vehicle, and the thermal system keeps the operating temperature range in check.

Unlike battery electric vehicles (BEVs) that require recharging from an external source, FCEVs produce electricity themselves and use a small and lightweight battery for storage. The main benefits of this propulsion technology include no engine noise, quick charging time of less than five minutes, and longer range as compared to fully electric vehicles, regard-